

Program : Diploma in Fire Technology and Safety	
Course Code: 3462	Course Title: Fire Engineering
Semester: 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:2 T:1 P:0)	Periods per semester: 45

Course Objectives:

- To understand the fundamentals of fire dynamics, combustion, and fire protection systems.
- To recognize the classification and applications of various types of fire.
- To analyze the performance of firefighting equipment and systems.
- To understand fire safety protocols and maintenance procedures.

Course Prerequisites:

Topic	Course code	Course Name	Semester
Principles of Safety Management		Principles of Safety Management	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify the construction and operation of fire systems and combustion.	10	Applying
CO2	Classify fire types and apply fire protection techniques.	11	Applying
CO3	Select and operate firefighting equipment and safety systems for specific scenarios	12	Applying
CO4	Choose methods for speed control and manage fire ground operations effectively.	10	Applying

	Series Test	2	
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CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2	2			
CO2	3	2	3	2			
CO3	3	2	3	3			
CO4	3	3	3	2			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand fire dynamics and combustion processes.		
M1.01	Illustrate the construction details of fire systems and machines.	3	Understanding
M1.02	Explain the principle of combustion and its effects on materials.	2	Understanding
M1.03	Analyze combustion equations and solve related problems.	3	Applying
M1.04	Compare different fire protection systems for specific scenarios.	2	Understanding

Contents: Fundamentals of Combustion

Temperature, heat, specific heat, flashpoint, fire point, ignition, combustion.
Types of combustion: rapid, spontaneous, explosions, smoldering combustion.
Flames and combustion zones: diffusion, premixed, and flame spread.
Hazards of combustion: harmful products, smoke, and fire gases.

CO2	Understand fire growth and heat transfer within enclosures.		
M2.01	Illustrate fire growth stages: ignition, flashover, decay.	2	Understanding
M2.02	Explain factors affecting heat release and the importance of design fire.	3	Understanding
M2.03	Apply methods to estimate energy release and heat transfer during fires.	3	Applying

M2.04	Analyze the effects of heat transfer in compartment fires.	3	Applying
	Series Test –1	1	
Contents: Fire Dynamics in Enclosures Fire growth in enclosures: ignition, flashover, decay stages. Heat release rate: calorific value, burning rate, and factors affecting heat release. Modes of heat transfer: convection and radiation. Flame height, plume characteristics, and ceiling jets			
CO3	Gain knowledge in fire service equipment operation and maintenance.		
M3.01	Identify the operation of fire pumps, hoses, and firefighting tools.	3	Understanding
M3.02	Illustrate the use of ladders and rescue tools in firefighting.	3	Applying
M3.03	Explain safety protocols during fire rescue operations	3	Understanding
M3.04	Outline maintenance procedures for fire equipment.	3	Applying
Contents: Fire Service Equipment and Tools Suction and delivery hoses, hose reels, fire pumps, and hydrants. Types of ladders, hydraulic platforms, and rescue tools. Small gear and miscellaneous equipment: ropes, lighting sets, and testing methods.			
CO4	Understand fire hydraulics, discharge calculations, and fire ground operations.		
M4.01	Understand pressure loss, flow in pipes and hoses, and nozzle reaction.	2	Understanding
M4.02	Illustrate water relay techniques for firefighting.	3	Applying
M4.03	Discuss the methods used in fire ground rescue and ventilation.	2	Understanding
M4.04	Analyze safety procedures for personnel during firefighting operations.	3	Applying
	Series Test – II	1	
Contents: Fire hydraulics: flow in pipes, fire hoses, nozzle reaction. Water relay techniques and estimation of water requirements. Fire ground operations: preplanning, rescue methods, ventilation, and salvage.			

Text / Reference:

T/R	Book Title/Author
R1	Ron Hirst, Underdowns, “Practical fire precautions”, Gower publishing company Ltd., England.
R2	Jain V.K. “Fire safety in buildings” (2nd edn.). New Age International (P) Ltd., New Delhi.
R3	Andrew L Simon, Fire hydraulics Prentice-Hall Inc., USA.
R4	Bjorn Karlsson and James G. Quintiere, Enclosure Fire Dynamics, CRC Press, USA.
R5	Dougal Drysdale An Introduction to Fire Dynamics, (2nd edn.), John Wiley & Sons Ltd.,England.